

Jerin Romijah Tuli

AI/ML Researcher — NLP, LLMs & Biomedical Signal Processing

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OBJECTIVE

Seeking a fully funded MS/PhD position in artificial intelligence and machine learning research, with a focus on natural language processing, LLM evaluation, and deep learning for biomedical signals. My work centers on making model outputs trustworthy under uncertainty — whether that means an LLM recognizing an unanswerable question in a retrieval-augmented pipeline, or a classifier correctly hedging on an ambiguous EEG epoch at a sleep-stage boundary.

EDUCATION

2022 - 2026	B.Sc. in Computer Science and Engineering Rajshahi University of Engineering & Technology (RUET), Bangladesh — CGPA 3.78/4.00 <i>Thesis: Automated EEG Sleep Stage Classification. Supervisor: Md. Nasif Osman Khansur.</i>
2020 - 2021	Higher Secondary Certificate (HSC), Science Viqarunnisa Noon School & College — GPA 5.00
2018 - 2019	Secondary School Certificate (SSC), Science Viqarunnisa Noon School & College — GPA 5.00

PUBLICATIONS (PEER-REVIEWED)

1. J. R. Tuli, M. S. A. Pritom, T. N. H. Naem. “MindFlayer at SemEval-2026 Task 8: DUALRAG — Answerability-Aware Generation for Multi-Turn RAG Conversations.” Proc. 20th International Workshop on Semantic Evaluation (SemEval 2026), ACL Anthology, pp. 2314-2321.
Zero-shot answerability-routing system built on Llama-4-Scout-17B (Groq API); ranked 8th of 26 teams with 100% refusal accuracy on unanswerable questions, outperforming a GPT-OSS-120B baseline.
2. J. R. Tuli, T. N. H. Naem, M. S. A. Pritom. “MindFlayer at SemEval-2026 Task 13: LACR-ENS — Calibration-Aware Ensemble Routing for Cross-Language AI-Generated Code Detection.” Proc. 20th International Workshop on Semantic Evaluation (SemEval 2026), ACL Anthology, pp. 2322-2329.
Calibration-aware ensemble (UniXCoder + GraphCodeBERT) for AI-generated code detection across 8 languages; cut out-of-distribution calibration error from 0.18 to 0.11 while improving macro-F1.

RESEARCH IN PROGRESS

Calibrated Answerability Gating for Multi-Turn RAG (CAG-MT). A confidence-gated retrieval-augmented generation system reaching 52.7% unanswerable recall at 88.6% answerable accuracy on MTRAG-842/MTRAG-UN-507, breaking a published 0.240 unanswerable-F1 ceiling. With M. S. A. Pritom.

The Social Mirage. A representation-level probing study of whether instruction-tuned LLMs retain private beliefs under social pressure; finds a persistent Representation-Mind Gap (+0.55) and a 7.7× rise in strategic conformity from base to instruction-tuned models. With M. S. A. Pritom.

Automated EEG Sleep Stage Classification. Undergraduate thesis: a deterministic spectral-enhancement module feeding per-channel vision transformers with gated cross-channel fusion; 78.77% Macro-F1 / 78.91% accuracy on a verified subject-independent split of Sleep-EDF Expanded. Being prepared for IEEE JBHI.

Bengali RAG Hallucination Evaluation Pipeline. Independent research isolating retrieval failure from generation failure in Bengali RAG; introduces a Language Gap Score against an English baseline. Targeting the MeLLM workshop, ACL 2026.

BanglishBench. 1,130-sample Bengali-English code-switching benchmark across safety classification, NLI, and QA, with a composite BBS metric. Accepted to SemEval 2026.

WORK EXPERIENCE

Mar 2025	Software Development Intern <i>Brain Station 23 Limited</i> Collaborated in an agile squad on AI-driven healthcare solutions, participating in daily scrums and code reviews; contributed to AI/ML model integration and production-grade software development under senior engineering mentors.
2025	System Analyst, Group Project <i>Gizan Tech</i> Analyzed organizational workflows and data-handling processes; designed DFDs and process models for centralized data management; translated findings into technical recommendations presented to stakeholders.

SKILLS & INTERESTS

Research Interests	Natural Language Processing, LLM Evaluation, Retrieval-Augmented Generation, Deep Learning, Biomedical Signal Processing
Languages	English — Full professional proficiency; Bangla — Native
Programming	Python, SQL, C, C++, Java, LaTeX
Platforms / Frameworks	PyTorch, TensorFlow/Keras, Groq API, Flask, Django, scikit-learn

CERTIFICATIONS

DataCamp — Data Analyst Associate
DataCamp — Data Manipulation with pandas
DataCamp — Intermediate Python
Machine Learning with Python — Completion Certificate
PHP (Laravel) Certificate — EDGE-DSTS-116-6631-00001
Brain Station 23 — Internship Certificate